

Wenshuo Wang

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EDUCATION

South China University of Technology, Guangzhou, China

Sep 2023 - Present

B.Eng, School of Future Technology, Major in Artificial Intelligence

- **GPA:** 3.6/4.0
- **Major Courses:** Fluid Mechanics, Analysis and Applications of Heat, Engineering Math: Calculus II, Engineering Math: Probability & Mathematical Statistics, Discrete Mathematics, Python, Data Structures, Machine Learning, Deep Learning and Computer Vision, Introduction to Cognitive Neuroscience, and related subjects
- **Languages:** English (TOEFL iBT 101)

Massachusetts Institute of Technology, Cambridge, MA, USA

Nov 2024 - Present

Research Intern (Remote), Kavli Institute for Astrophysics and Space Research (Advisor: Prof. [Fan Zhang](#))

RESEARCH INTERESTS

My research focuses on **deep learning for scientific computing**, especially **neural PDE simulation** for physical-field dynamics. I study neural PDE simulators beyond architecture design, focusing on how problem states, observations, representations, and computational budgets determine rollout fidelity and physical reliability. My long-term goal is to develop efficient, reliable, and physically faithful neural simulation methods for scientific and engineering systems.

PUBLICATIONS AND PREPRINTS¹

[1] Posterior-First Neural PDE Simulation: Inferring Hidden Problem State from a Single Field

Preprint [\[Paper\]](#)

Wenshuo Wang, Fan Zhang[†] (Corresponding Author)*

- Theoretically establish that a single observed field can leave multiple task-relevant problem states unresolved, making posterior problem-state inference necessary before reliable neural PDE rollout, rather than deterministic field-to-future prediction.
- Introduce a posterior-first simulator that improves rollout fidelity and hidden-physics recovery.

[2] Derived Fields Preserve Fine-Scale Detail in Budgeted Neural Simulators

Preprint [\[Paper\]](#)

Wenshuo Wang, Fan Zhang[†]

- Reposition carried-state design as the primary bottleneck in budgeted neural simulation by showing that, under tight storage budgets, fine-scale fidelity is often determined when the state is constructed, before stronger architectures, losses, or rollout strategies can recover information already discarded.
- Propose DerivOpt, a budgeted state-design framework that optimizes which physical fields a neural simulator carries and how storage is allocated across them.

[3] Breaking Scale Anchoring: Frequency Representation Learning for Accurate High-Resolution Inference from Low-Resolution Training

ICLR 2026 [\[Paper\]](#)

Wenshuo Wang, Fan Zhang[†]

- Identify Scale Anchoring and propose Frequency Representation Learning for more accurate high-resolution spatiotemporal inference from low-resolution training.
- Correct the field's notion of multi-resolution generalization by showing that stable error across resolutions can mask a deeper failure to learn the true physical operator.

¹ Grouped by area: spatiotemporal forecasting, knowledge distillation, and LLM/VLM.

[4] Gradient-Informed Temporal Sampling Improves Rollout Accuracy in PDE Surrogate Training

Preprint [\[Paper\]](#)

Wenshuo Wang, Fan Zhang[†]

- Propose GITS, a temporal sampling method that combines pilot-model gradient scores with set-level temporal coverage for PDE surrogate training.

[5] System-Anchored Knee Estimation for Low-Cost Context Window Selection in PDE Forecasting

Preprint [\[Paper\]](#)

Wenshuo Wang, Fan Zhang[†]

- Propose SAKE, a two-stage low-cost method that uses system-derived anchors and knee-aware search to select context windows for autoregressive neural PDE simulators.

[6] S²-KD: Semantic-Spectral Knowledge Distillation Spatiotemporal Forecasting

AAAI 2026 [\[Paper\]](#)

Wenshuo Wang, Yaomin Shen, Yingjie Tan, Yihao Chen[†]

- Propose S²-KD, a semantic-spectral distillation framework that transfers causal semantics and spectral knowledge from a multimodal teacher to a vision-only student for spatiotemporal forecasting.

[7] Knowledge Distillation Must Account for What It Loses

Preprint [\[Paper\]](#)

Wenshuo Wang

- Argue that knowledge distillation should account for teacher-capability losses rather than merely infer capability preservation from benchmark-score retention.

[8] LLM Reasoning Is Latent, Not the Chain of Thought

Preprint [\[Paper\]](#) [Featured by JIQIZHIXIN](#)

Wenshuo Wang

- Argue that LLM reasoning should, by default, be studied as latent-state trajectory formation rather than as faithful surface chain-of-thought.

[9] iTAG: Inverse Design for Text Generation with Accurate Causal Graph Annotations

ACL 2026 (Oral Presentation) [\[Paper\]](#)

Wenshuo Wang, Boyu Cao, Nan Zhuang, Wei Li[†]

- Propose iTAG, an inverse-design framework that assigns real-world concepts to causal graphs and generates natural text with highly accurate causal annotations.

[10] Pi-CCA: Prompt-Invariant CCA Certificates for Replay-Free Vision-Language Continual Learning

ICLR 2026 [\[Paper\]](#)

Jiayu Zhang, Chuangxin Zhao, Ruibo Duan, Canran Xiao, Wenyi Mo, Haoyu Gao, Wenshuo Wang[†]

- Introduce PI-CCA, a replay-free continual multimodal learning method that preserves image-text alignment geometry with compact CCA certificates.

INTERNSHIP EXPERIENCE

HUAWEI Beijing Research Institute, Beijing, China

Jul 2024 - Oct 2024

Algorithm Engineer Intern

- Developed NLP and medical AI applications for text understanding, summarization, retrieval, and conversational question answering.
- Conducted research on NLP, focusing on LLMs.